HANKEL DETERMINANTS OF DISCRETE MEASURES WITH POWER-LAW ATOMS, AND OF ZETA VALUES

GORDON [SURNAME]

ABSTRACT. Let $\sigma=\sum_{k\ge1}w_k\delta_{x_k}$ be a discrete measure with atoms $x_k=f(k)$, where $f'(\xi)\sim -pC\xi^{-p-1}$ for some $p>0$ and $C>0$ (so that $x_k-\lim_j x_j\sim Ck^{-p}$), and with weights satisfying $\sum_k w_k<\infty$ and $w_k\ge c_1k^{-s}$. We prove that the Hankel determinants $D_n=\det(\int x^{i+j}\,\mathrm{d}\sigma)_{i,j=0}^{n-1}$ satisfy

$$\log D_n=-pn^2\log n+\Big(\tfrac{3p}{2}+\log\big(c(p)C\big)\Big)n^2+o(n^2),\qquad c(p)=\tfrac14\Big(\tfrac{\sqrt{\pi}\,\Gamma(1/p+1)}{\Gamma(1/p+1/2)}\Big)^p.$$

Equivalently, a double Cesàro mean of $\log(j^pb_j)$, where the b_j are the off-diagonal Jacobi coefficients of σ, converges to $\log(c(p)C)$. The constant comes from an explicitly solvable constrained equilibrium problem: the minimiser of the logarithmic energy

$$\iint\log\frac{1}{|u^{-p}-v^{-p}|}\,\mathrm{d}\mu(u)\,\mathrm{d}\mu(v)$$

among probability measures $\mu\le \mathrm{d}u$ on $(0,\infty)$ is unique, it is an average of arcsine laws, and its density is a regularised incomplete beta function. As a corollary, for every real $a>0$ and real b with $2a+b>1$, $\log\det_{1\le i,j\le n}\zeta(a(i+j)+b)=-an^2\log n+(\frac{3a}{2}+\log c(a))n^2+o(n^2)$. For positive integers a,b this gives the leading-order behaviour $-an^2\log n$ expected by Haynes and Zudilin, together with the n^2 term. The main theorem and both corollaries have been formalised in the Lean 4 proof assistant.

1. INTRODUCTION

1.1. **Setting and main result.** For a positive Borel measure σ on $\mathbb{R}$ with finite moments $m_r=\int x^r\,\mathrm{d}\sigma(x)$, write

$$D_n(\sigma):=\det\big(m_{i+j}\big)_{i,j=0}^{n-1}\qquad(n\ge1)$$

for its Hankel determinants. We study discrete measures

$$\sigma=\sum_{k\ge1}w_k\,\delta_{x_k},\qquad x_k=f(k),\tag{1.1}$$

whose atoms accumulate at a single point with polynomial spacing, the model case being $x_k=Ck^{-p}$. Throughout, $p>0$ and

$$I(p):=\frac{\sqrt{\pi}}{2}\,\frac{\Gamma(\frac1p+\frac12)}{\Gamma(\frac1p+1)},\qquad c(p):=\frac14\Big(\frac{\pi}{2I(p)}\Big)^p,\qquad \kappa_p:=\frac{3p}{2}+\log c(p).\tag{1.2}$$

For instance $I(1)=\pi/4$, $c(1)=\frac12$, and $I(2)=1$, $c(2)=\pi^2/16$.

Theorem 1.1. *Let $p>0$ and $C>0$. Let $f\in C^1[1,\infty)$ satisfy*

$$\lim_{\xi\to\infty}\xi^{p+1}f'(\xi)=-pC,\qquad 0<\inf_{\xi\ge1}\xi^{p+1}|f'(\xi)|\le\sup_{\xi\ge1}\xi^{p+1}|f'(\xi)|<\infty,\tag{1.3}$$

Date: 15 October 2026.

2020 *Mathematics Subject Classification.* Primary 42C05; Secondary 15B05, 31A15, 11M06, 47B36.

Key words and phrases. Hankel determinants, discrete orthogonal polynomials, Jacobi matrices, constrained equilibrium problem, logarithmic energy, arcsine law, Riemann zeta function, formal verification.

and let the weights $w_k \geq 0$ *satisfy*

$$W := \sum_{k\geq 1} w_k < \infty \qquad \text{and} \qquad w_k \geq c_1 k^{-s} \quad (k \geq 1) \tag{1.4}$$

for some constants $c_1 > 0$ *and* $s \in \mathbb{R}$. *Then the measure* σ *of* (1.1) *satisfies*

$$\log D_n(\sigma) = -pn^2 \log n + \big(\kappa_p + \log C\big)n^2 + o(n^2) \qquad (n \to \infty). \tag{1.5}$$

The upper bound in the second condition of (1.3) already follows from the limit and the continuity of f'; we state it for symmetry. Under (1.3), f is strictly decreasing with a finite limit L, and $f(\xi) - L \sim C\xi^{-p}$; Hankel determinants are invariant under translation of σ (Lemma 2.2), so L plays no role. The upper bound in (1.5) uses only $\sum_k w_k < \infty$; apart from the finiteness of the moments, the lower bound uses only $w_k \geq c_1 k^{-s}$.

Since σ has infinitely many atoms, it has monic orthogonal polynomials of all degrees and a Jacobi matrix with off-diagonal entries $b_j > 0$ (Lemma 2.4). Theorem 1.1 is equivalent to a doubly averaged statement about these coefficients.

Corollary 1.2. *Under the hypotheses of Theorem 1.1,*

$$\frac{2}{n^2}\sum_{j=1}^{n-1}(n-j)\log\big(j^p b_j\big) \longrightarrow \log\big(c(p)C\big).$$

The constant κ_p is the value of a variational problem, which we solve explicitly. Let

$$\mathcal{A} := \big\{\mu \text{ positive Borel measure on } (0,\infty) :\ \mu\big((0,\infty)\big) = 1,\ \mu \leq \mathrm{Leb}\big\},$$

where $\mu \leq \mathrm{Leb}$ means $\mu(B) \leq |B|$ for every Borel set B, and for $\mu \in \mathcal{A}$ put

$$\mathcal{J}_p(\mu) := \iint \log\big|u^{-p} - v^{-p}\big|\,\mathrm{d}\mu(u)\,\mathrm{d}\mu(v) \in [-\infty, \infty). \tag{1.6}$$

For $\beta > 0$ let ω_β be the arcsine probability measure on $[0,\beta]$, with density $1/(\pi\sqrt{y(\beta-y)})$. For $V \in [0,1]$ and $\alpha, \gamma > 0$ let $\mathcal{B}_V(\alpha,\gamma) := B(\alpha,\gamma)^{-1}\int_0^V s^{\alpha-1}(1-s)^{\gamma-1}\,\mathrm{d}s$ be the regularised incomplete beta function.

Theorem 1.3. *Let* $p > 0$ *and* $c = c(p)$. *Let* μ^* *be the probability measure on* $(0,\infty)$

$$\mu^* := \int_0^1 \omega_{4ct^{-p}}\,\mathrm{d}t, \qquad \textit{i.e.} \quad \mu^*(B) = \int_0^1 \omega_{4ct^{-p}}(B)\,\mathrm{d}t,$$

and let μ_u^* *be its image under* $x \mapsto x^{-1/p}$. *Then:*

1. $\mu_u^* \in \mathcal{A}$, *with density* $\rho^*(u) = \mathcal{B}_{\min(1,\,u^{-p}/4c)}\big(\frac{1}{p}+\frac{1}{2},\frac{1}{2}\big)$. *In particular* $\rho^* = 1$ *on* $(0,u_s]$, *where* $u_s := (4c)^{-1/p} = 2I(p)/\pi$, *and* $0 < \rho^* < 1$ *on* (u_s,∞).
2. *The function* $U^*(x) := \int \log|x-y|\,\mathrm{d}\mu^*(y)$ *(the negative of the logarithmic potential of* μ^**) satisfies* $U^*(x) = \log c + p$ *for* $0 < x \leq 4c$, *and* $\geq \log c + p$ *for all* $x > 0$.
3. $\mathcal{J}_p(\mu_u^*) = \kappa_p$, *and* $\mathcal{J}_p(\mu) < \kappa_p$ *for every* $\mu \in \mathcal{A}$ *with* $\mu \neq \mu_u^*$. *Thus* $\sup_{\mathcal{A}} \mathcal{J}_p = \kappa_p$, *attained only at* μ_u^*.

Applied to the measure $\sum_k k^{-(2a+b)}\delta_{k^{-a}}$, Theorem 1.1 gives the asymptotics of Hankel determinants of zeta values along arithmetic progressions.

Corollary 1.4. *Let* $a > 0$ *and* b *be real numbers with* $2a + b > 1$. *Then*

$$\log \det_{1\leq i,j\leq n} \zeta\big(a(i+j)+b\big) = -an^2\log n + \Big(\frac{3a}{2} + \log c(a)\Big)n^2 + o(n^2).$$

In particular, $\log\det_{1\le i,j\le n}\zeta(i+j+b) = -n^2\log n + (\frac{3}{2}-\log 2)n^2 + o(n^2)$ *for every* $b>-1$, *and* $\log\det_{1\le i,j\le n}\zeta(2(i+j)+b) = -2n^2\log n + (3+\log\frac{\pi^2}{16})n^2 + o(n^2)$ *for every* $b>-3$.

1.2. **Context.** *Hankel determinants of zeta values.* Monien [12] studied the Hankel determinants $H_n^{(r)}[F] = \det(F(i+j+r))$ of Dirichlet series F, gave a multiple-sum formula for them [12, Thm. 2.1], and wrote for $F=\zeta$: "We do not know how to prove the full asymptotic expansions but will describe a method that lets one at least understand the weaker asymptotic form $\log H_n^{(0)}[\zeta] \sim \log H_n^{(1)}[\zeta] \sim -n^2(\log(2n)-3/2)$." He also reports numerical experiments of D. Zagier consistent with this form. Haynes and Zudilin [8] wrote: "for positive integers a and b, we expect that $\log\det_{1\le i,j\le n}\zeta(a(i+j)+b) = -an^2\log n + O(n^2)$ as $n\to\infty$", and proved the weaker bound "$\log\det_{1\le i,j\le n}\zeta(a(i+j)+b) \le -(a/2)n^2\log n + O(n^2)$". They also write that "In [5] Monien developed these ideas further and rigorously justified the above asymptotics in the case $r=0$", their reference [5] being a personal communication [8, §1]. Coons and Sutherland [3, Thm. 1.3] proved, for every nondegenerate Dirichlet series $F(s)=\sum f(n)n^{-s}$ with $f\ge 0$ convergent for $\Re s \ge s_0$, that $H_n^{(r)}[F]>0$ for $n\ge 1$, $r\ge s_0-2$, and that $\log H_n^{(r)}[F] < -cn^2$ for some $c>0$ and all sufficiently large n and r.

Corollary 1.4 therefore proves the asymptotic formula expected in [8], for all positive integers a,b and more generally for real $a>0$ and $2a+b>1$, and it identifies the n^2 term. For $a=1$ the constant $\frac{3}{2}-\log 2$ agrees with Monien's form $-n^2(\log 2n - \frac{3}{2}) = -n^2\log n + (\frac{3}{2}-\log 2)n^2$, provided the latter is read with an $o(n^2)$ error, as the expansion from Zagier's numerics quoted in [12] indicates. (Read literally, Monien's "$\sim$" carries only the leading term.) For $a=2$ and $b\in\{-2,0,2\}$ the determinants are known in closed form, by results of Baricz and Štampach [1] on Hankel determinants of Rayleigh functions (the closed forms we use were confirmed independently against direct determinants); their asymptotics are consistent with Corollary 1.4 (Remark 7.2). Apart from these closed-form cases, we have not found in the literature a published proof of the leading term $-an^2\log n$ (for $a=1$, $r=0$ see the remark of [8] quoted above), nor of the n^2 term $\frac{3a}{2}+\log c(a)$.

Hankel determinants of zeta-type moments also occur in recent irrationality work. Lai [11] writes that "Fauzan's proof uses Hankel determinants, which were previously studied by Haynes–Zudilin [23] and Zudilin [55]". Fauzan's announcement [7] is an unrefereed Zenodo record; the full manuscript was not available to us. Its record page describes "normalized Hankel determinants" together with "a positive moment representation and a logarithmic-energy estimate". The determinants there appear to be built from a different moment sequence. There is nonetheless a methodological overlap with the present paper (Hankel determinants of zeta-type moments controlled by a logarithmic-energy argument), and we cannot exclude that the announced energy estimate is related to Theorem 1.3. The present paper is not concerned with arithmetic applications, and none of its proofs uses [7] or [11].

The equilibrium problem. Since $-\mathcal{J}_p(\mu)$ is the logarithmic energy of the image measure $g_*\mu$, the maximisation of $\mathcal{J}_p$ over $\mathcal{A}$ is a logarithmic energy minimisation with an upper constraint on the measure. Constrained energy problems of this kind are the subject of a substantial literature. For the following works we have consulted only the abstracts, and the summary below is restricted to what those abstracts say. According to the abstract of [4], the constrained equilibrium problem was introduced by Rakhmanov [13] to describe the asymptotic zero distribution of discrete orthogonal polynomials in a compact interval, and was extended by Dragnev and Saff [6] and by Kuijlaars and Van Assche [10]. The abstract of [6] announces existence and uniqueness of the constrained minimiser, a characterisation theorem and a saturation result, with explicit results for Krawtchouk polynomials. The abstract of [10] states that for extremal polynomials on infinite discrete sets such as those of the Meixner and Charlier polynomials,

the asymptotics are governed by a constrained extremal energy problem that can be solved explicitly. The abstract of Van Assche and Van Baelen [14] concerns a q-lattice $\{q^k\}$, which accumulates at 0. We use no result from this theory: existence, uniqueness and the value of the extremum are proved directly in Section 3. We have not found the problem $\sup_{\mathcal{A}} \mathcal{J}_p$, whose constraint has infinite total mass near the accumulation point, or its solution, in the literature. Since we have not read the full texts of [13, 6, 10, 9], this statement is provisional.

For $p = 1$, Monien [12, §3] set up a constrained maximisation of this type heuristically: "We seek a continuous test function $\rho(x)$ which maximizes the functional Eq. 3.5 subject to the constraint that $0 < \rho(x) \le 1$ for all x and $\rho(x) = 0$ for $x < 0$ and Eq. 3.4. We assume that such a continuous functions exists." His Theorem 3.1 gives the solution of the stationarity condition,

$$\rho(x) = 1 \quad (0 \le x \le \tfrac{1}{2}), \qquad \rho(x) = \tfrac{2}{\pi}\left(\arctan \tfrac{1}{\sqrt{2x-1}} - \tfrac{\sqrt{2x-1}}{2x}\right) \quad (x > \tfrac{1}{2}),$$

which coincides with the density ρ^* of Theorem 1.3 for $p = 1$ (Remark 3.6). His argument assumes existence, checks only stationarity and does not contain an upper bound or a discretisation step. Theorem 1.3 proves optimality and uniqueness for every $p > 0$, and Sections 4–5 supply the passage from the continuous problem to the determinants.

Averaged arcsine laws. The measure μ^* is an average of arcsine laws. A theorem of Kuijlaars and Van Assche on recurrence coefficients that vary regularly with n/N ([9], quoted from [5, Thm. 1.3]) produces measures of exactly this shape as limiting zero distributions. With their conventions, a family of Jacobi matrices whose diagonal and off-diagonal entries (after rescaling by N^p) behave like $2c\,t^{-p}$ and $c\,t^{-p}$ as $n/N \to t$ has, at $t = 1$, the limiting zero distribution $\int_0^1 \omega_{[0,4ct^{-p}]}\,\mathrm{d}t$. Heuristically, therefore, μ^* is the zero distribution that the measures of Theorem 1.1 would have if their recurrence coefficients satisfied $a_n \sim 2b_n \sim 2c(p)Cn^{-p}$ pointwise; the saturation condition in Theorem 1.3(i) is exactly the equation that fixes $c = c(p)$. This heuristic is not used in the proofs, and we do not prove pointwise asymptotics of the recurrence coefficients: Corollary 1.2 is a statement about double Cesàro means only.

1.3. **Method.** By Heine's formula (Lemma 2.1), D_n is a sum over n-subsets S of atoms of $\prod_{k\in S} w_k$ times a squared Vandermonde product. For $x_k = k^{-p}$ and $S = \{k_1 < \cdots < k_n\}$, the Vandermonde product is $n^{-pn(n-1)} \prod_{i\neq j} |g(u_i) - g(u_j)|$ with $g(u) = u^{-p}$ and $u_i = k_i/n$, and the empirical measure $\frac{1}{n}\sum \delta_{u_i}$ has at most one atom in each interval of length $1/n$, a discrete form of the constraint $\mu \le \mathrm{Leb}$. The upper bound (Section 4) compares point configurations with step densities (Lemma 4.1) and removes atoms beyond Mn by a tail estimate; the lower bound (Section 5) evaluates a single term of the Heine sum at the quantiles of a truncation of μ_u^*. Both bounds meet at $\sup_{\mathcal{A}} \mathcal{J}_p$, which Theorem 1.3 identifies. General regularly spaced atoms are reduced to $x_k = Ck^{-p}$ by a mean-value comparison (Lemma 2.3).

The only potential-theoretic input beyond elementary calculus is the positivity of the logarithmic energy of signed measures of total mass zero (Lemma 3.3), which is used only for the optimality and uniqueness part of Theorem 1.3(iii). It is a special case of a lemma of Beckermann, Kalyagin, Matos and Wielonsky [2, Lemma 2.1]; we include a short Fourier-analytic proof so that the paper is self-contained (Remark 3.4).

1.4. **Formal verification.** Theorem 1.1, Corollary 1.2 and Corollary 1.4 have been formalised in the Lean 4 proof assistant (version 4.34.1) with the Mathlib library, with the hypotheses as stated above (Lean theorems `paper1_thm11`, `paper1_cor12` and `paper1_cor14`; in Corollary 1.4, ζ is Mathlib's Riemann zeta function). The formalisation contains no unproved steps and uses only Lean's standard axioms (`propext`, `Quot.sound`, `Classical.choice`). It also

covers the parts of Theorem 1.3 that the proof of Theorem 1.1 uses: admissibility of μ_u^* and its saturation on $(0, u_s]$ (but not the incomplete-beta formula for ρ^*), the potential of μ^* (part (ii)), the value $\mathcal{J}_p(\mu_u^*) = \kappa_p$, the inequality $-\iint \log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu \geq 0$ of Lemma 3.3 (not the Fourier identity and not the equality case), and the bound $\mathcal{J}_p(\mu) \leq \kappa_p$ for $\mu \in \mathcal{A}$ with bounded support and integrable energy, which is the case the proof uses. Uniqueness of the maximiser is not formalised. In a few places the formal proof takes a different route from the one written here; for example, it proves Lemma 3.2 with log-sine integrals instead of Fourier series, and Lemma 3.3 with a Gaussian kernel instead of the Fourier transform. The Lean sources are part of the supplementary material (Section 8).

1.5. **Organisation and notation.** Section 2 collects the facts about Hankel determinants that we use. Section 3 proves Theorem 1.3. Sections 4 and 5 prove the upper and lower bounds for $x_k = k^{-p}$, and Section 6 completes the proofs of Theorem 1.1 and Corollary 1.2. Section 7 treats zeta values, and Section 8 reports numerical illustrations.

Throughout, $p > 0$ is fixed and $g(u) := u^{-p}$ on $(0, \infty)$. Constants denoted C_M, C_U, ... depend only on p and on the indicated parameters, and may change from line to line. We use the convention $0 \log 0 = 0$.

2. Hankel determinants of discrete measures

In this section $\sigma = \sum_{k \geq 1} w_k \delta_{x_k}$ with $w_k \geq 0$, $\sum_k w_k < \infty$ and $\sup_k |x_k| < \infty$, so that all moments are finite. For a finite set $S \subset \mathbb{N}$ write

$$w(S) := \prod_{k \in S} w_k, \qquad V_x(S) := \prod_{\substack{k,l \in S \\ k<l}} (x_k - x_l)^2.$$

Lemma 2.1 (Heine's formula)**.** *For every $n \geq 1$,*

$$D_n(\sigma) = \sum_{\substack{S \subset \mathbb{N} \\ |S|=n}} w(S)\, V_x(S).$$

In particular all terms are nonnegative.

Proof. Write $m_{i+j} = \sum_k w_k x_k^i x_k^j$. Expanding the determinant multilinearly in its columns (legitimate because $\sum_k w_k |x_k|^r \leq W X^r$ with $W = \sum_k w_k$ and $X := \sup_k |x_k|$, so all series converge absolutely),

$$D_n = \sum_{k_0, \dots, k_{n-1} \geq 1} \Big(\prod_{j=0}^{n-1} w_{k_j} x_{k_j}^j \Big) \det\big(x_{k_j}^i\big)_{i,j=0}^{n-1}.$$

The Vandermonde determinant vanishes unless the k_j are distinct. Group the remaining tuples according to the set $S = \{l_0 < \cdots < l_{n-1}\}$ they enumerate, $k_j = l_{\pi(j)}$ with π a permutation. Then $\det(x_{k_j}^i) = \operatorname{sgn}(\pi) \det(x_{l_j}^i)$ and

$$\sum_{\pi} \operatorname{sgn}(\pi) \prod_j x_{l_{\pi(j)}}^j = \det\big(x_{l_i}^j\big),$$

so the contribution of S is $w(S) \det(x_{l_j}^i)^2 = w(S) V_x(S)$. □

Lemma 2.2. *Let σ, σ' be measures as above.*

1. *(Monotonicity) If $\sigma' = \sum w'_k \delta_{x_k}$ with $w_k \leq w'_k$ for all k, then $D_n(\sigma) \leq D_n(\sigma')$.*
2. *(Translation) $D_n(\sum w_k \delta_{x_k + t}) = D_n(\sigma)$ for every $t \in \mathbb{R}$.*

(3) *(Scaling)* $D_n(\sum w_k\delta_{\lambda x_k}) = \lambda^{n(n-1)}D_n(\sigma)$ *for* $\lambda > 0$, *and* $D_n(\sum \vartheta w_k\delta_{x_k}) = \vartheta^n D_n(\sigma)$ *for* $\vartheta > 0$.
(4) *(Positivity) If* $w_k > 0$ *for infinitely many* k *with pairwise distinct* x_k, *then* $D_n(\sigma) > 0$ *for all* n.

Proof. All four statements follow from Lemma 2.1: its terms are nonnegative and increasing in each w_k; $V_x(S)$ depends only on differences; $V_{\lambda x}(S) = \lambda^{n(n-1)}V_x(S)$ and $(\vartheta w)(S) = \vartheta^n w(S)$; and under the hypothesis of (iv) some term is positive. □

Lemma 2.3 (Regular spacing comparison)**.** *Let* $f, h \in C^1[1,\infty)$ *with* $f' < 0$ *and* $h' < 0$ *on* $[1,\infty)$, *both bounded on* $[1,\infty)$, *and suppose that* $\rho := f'/h'$ *satisfies* $R^{-1} \le \rho \le R$ *on* $[1,\infty)$ *for some* $R \ge 1$ *and* $\rho(\xi) \to 1$ *as* $\xi \to \infty$. *Put* $\eta(m) := \sup_{\xi\ge m} |\log \rho(\xi)|$. *Then for all weights* $w_k \ge 0$ *with* $\sum w_k < \infty$ *and all* $n \ge 1$,

$$\Big|\log D_n\Big(\sum_k w_k\delta_{f(k)}\Big) - \log D_n\Big(\sum_k w_k\delta_{h(k)}\Big)\Big| \le 2\sum_{i=1}^{n-1}(n-i)\,\eta(i) = o(n^2),$$

whenever one (hence both) of the determinants is positive.

Proof. Let $S = \{k_1 < \cdots < k_n\}$. For $i < j$, Cauchy's mean value theorem gives $(f(k_i) - f(k_j))/(h(k_i) - h(k_j)) = \rho(\xi)$ for some $\xi \in (k_i, k_j)$. Since $\xi > k_i \ge i$, the ratio of the squared factors lies in $[e^{-2\eta(i)}, e^{2\eta(i)}]$. Multiplying over pairs, $V_f(S)/V_h(S)$ lies in $[e^{-E_n}, e^{E_n}]$ with $E_n = 2\sum_{i<n}(n-i)\eta(i)$, and the bound follows by comparing the Heine sums term by term. Finally η is bounded by $\log R$ and tends to 0, so $E_n = o(n^2)$. □

Lemma 2.4 (Jacobi coefficients)**.** *Suppose* $D_n(\sigma) > 0$ *for all* n. *Let* P_k *be the monic orthogonal polynomials of* σ *and* $h_k := \int P_k^2\,\mathrm{d}\sigma$. *Then* $D_n = \prod_{k=0}^{n-1} h_k$. *Moreover* $xP_k = P_{k+1} + a_{k+1}P_k + b_k^2 P_{k-1}$ *for* $k \ge 1$, *with* $b_k := (h_k/h_{k-1})^{1/2} > 0$, *and*

$$D_n(\sigma) = m_0^n \prod_{j=1}^{n-1} b_j^{2(n-j)}.$$

The numbers b_k are the off-diagonal entries of the Jacobi matrix of σ (the coefficients of the three-term recurrence for the orthonormal polynomials $P_k/\sqrt{h_k}$).

Proof. The Hankel matrix $H_n = (m_{i+j})_{i,j<n}$ is the Gram matrix of $1, x, \ldots, x^{n-1}$ in $L^2(\sigma)$. Writing $(P_0, \ldots, P_{n-1})^T = L(1, \ldots, x^{n-1})^T$ with L unit lower triangular gives $LH_nL^T = \mathrm{diag}(h_0, \ldots, h_{n-1})$, so $D_n = \prod_{k<n} h_k$. The polynomial $xP_k - P_{k+1}$ has degree $\le k$, and it is orthogonal to P_l for $l \le k-2$; its P_{k-1}-coefficient is

$$\frac{1}{h_{k-1}}\int xP_kP_{k-1}\,\mathrm{d}\sigma = \frac{1}{h_{k-1}}\int P_k\,(xP_{k-1})\,\mathrm{d}\sigma = \frac{h_k}{h_{k-1}}.$$

Hence $h_k = m_0 b_1^2 \cdots b_k^2$ and $D_n = \prod_{k=0}^{n-1} m_0 \prod_{j\le k} b_j^2 = m_0^n \prod_{j=1}^{n-1} b_j^{2(n-j)}$. □

Lemma 2.5. $2\sum_{j=1}^{n-1}(n-j)\log j = n^2\log n - \frac{3}{2}n^2 + O(n\log n)$.

Proof. Let $\phi(x) = (n-x)\log x$ on $[1,n]$. Then $|\phi'(x)| \le \log n + n/x$, so $\big|\sum_{j=1}^{n-1}\phi(j) - \int_1^n \phi\big| \le \sum_{j=1}^{n-1}(\log n + n/j) = O(n\log n)$, and $\int_1^n (n-x)\log x\,\mathrm{d}x = \frac{n^2}{2}\log n - \frac{3}{4}n^2 + O(n)$. □

3. The constrained equilibrium problem

3.1. **Energies.** For finite positive Borel measures μ, ν on $(0,\infty)$ with $\int \log(1+x)\,\mathrm{d}\mu < \infty$ and $\int \log(1+x)\,\mathrm{d}\nu < \infty$ put

$$\mathcal{L}(\mu,\nu) := \iint \log|x-y|\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y), \qquad \mathcal{L}(\mu) := \mathcal{L}(\mu,\mu).$$

Since $\log|x-y| \le \log(1+x)+\log(1+y)$ for $x,y>0$, the positive part of the integrand is integrable, so $\mathcal{L}(\mu,\nu) \in [-\infty,\infty)$ is well defined. Let

$$\sigma_p(\,\mathrm{d}x) := \tfrac{1}{p}\,x^{-1-1/p}\,\mathrm{d}x \quad \text{on } (0,\infty),$$

the image of Lebesgue measure on $(0,\infty)$ under $g(u)=u^{-p}$. For $\tau>0$ let $\mathcal{A}_\tau$ be the set of positive Borel measures on $(0,\infty)$ of mass τ with $\mu \le \mathrm{Leb}$; thus $\mathcal{A} = \mathcal{A}_1$. For $\mu \in \mathcal{A}_\tau$ the image $\mu^x := g_*\mu$ satisfies $\mu^x \le \sigma_p$, hence $\int \log(1+x)\,\mathrm{d}\mu^x \le \tau\log 2 + \int_1^\infty \log(1+x)\,\mathrm{d}\sigma_p < \infty$, and we define

$$\mathcal{J}_p(\mu) := \mathcal{L}(\mu^x) = \iint \log|g(u)-g(v)|\,\mathrm{d}\mu(u)\,\mathrm{d}\mu(v) \in [-\infty,\infty), \qquad E_p := \sup_{\mu\in\mathcal{A}} \mathcal{J}_p(\mu).$$

Lemma 3.1 (Scaling)**.** *For a measure μ on $(0,\infty)$ and $\tau>0$ let $\mu_\tau(B) := \tau\mu(B/\tau)$. If μ has mass m and density $\rho \le 1$, then μ_τ has mass τm and density $\rho(\cdot/\tau) \le 1$, and $\mathcal{J}_p(\mu_\tau) = \tau^2(\mathcal{J}_p(\mu) - pm^2\log\tau)$. In particular $\mu \mapsto \mu_\tau$ is a bijection $\mathcal{A} \to \mathcal{A}_\tau$ and $\sup_{\mathcal{A}_\tau}\mathcal{J}_p = \tau^2(E_p - p\log\tau)$.*

Proof. $\mu_\tau(B) = \tau\int_{B/\tau}\rho(u)\,\mathrm{d}u = \int_B \rho(v/\tau)\,\mathrm{d}v$. Since $g(\tau u) = \tau^{-p}g(u)$,

$$\mathcal{J}_p(\mu_\tau) = \tau^2 \iint \log|g(\tau u)-g(\tau v)|\,\mathrm{d}\mu\,\mathrm{d}\mu = \tau^2\big(\mathcal{J}_p(\mu) - pm^2\log\tau\big).$$

□

3.2. **Arcsine potentials.**

Lemma 3.2. *Let $\beta>0$ and $U_\beta(x) := \int \log|x-y|\,\mathrm{d}\omega_\beta(y)$. Then $U_\beta(x) = \log(\beta/4)$ for $x \in [0,\beta]$, and U_β is increasing on $[\beta,\infty)$; in particular $U_\beta \ge \log(\beta/4)$ on $[0,\infty)$. Moreover, for every $x \ge 0$,*

$$\int \big|\log|x-y|\big|\,\mathrm{d}\omega_\beta(y) \le 2\log(1+x+\beta) + \big|\log(\beta/4)\big|. \tag{3.1}$$

Proof. Write $x = \frac{\beta}{2}(1-\cos\varphi)$ and $y = \frac{\beta}{2}(1-\cos\theta)$ with $\theta,\varphi \in [0,\pi]$. Under $y = \frac{\beta}{2}(1-\cos\theta)$, the measure ω_β is the image of $\mathrm{d}\theta/\pi$ on $[0,\pi]$, since $\sqrt{y(\beta-y)} = \frac{\beta}{2}\sin\theta$. Now $\log|x-y| = \log\frac{\beta}{4} + \log|2(\cos\theta-\cos\varphi)|$, and

$$\log|2(\cos\theta-\cos\varphi)| = \log\Big|2\sin\tfrac{\theta+\varphi}{2}\Big| + \log\Big|2\sin\tfrac{\theta-\varphi}{2}\Big| = -\sum_{k\ge1}\frac{2}{k}\cos k\theta\cos k\varphi,$$

using $\log|2\sin(\alpha/2)| = -\sum_{k\ge1}k^{-1}\cos k\alpha$, the Fourier series of an $L^2(0,2\pi)$ function. Each of the two series converges in $L^2([0,\pi],\mathrm{d}\theta)$, hence in L^1, so we may integrate termwise; since $\int_0^\pi \cos k\theta\,\mathrm{d}\theta = 0$ for $k\ge1$, we get $U_\beta(x) = \log(\beta/4)$ for $x\in[0,\beta]$. For $x>\beta$, $U_\beta'(x) = \int (x-y)^{-1}\,\mathrm{d}\omega_\beta(y) > 0$, and U_β is continuous at β by dominated convergence (near $y=\beta$ the integrand is dominated by a multiple of $|\log s|\,s^{-1/2}$, $s=\beta-y$). For (3.1), write $|t| = 2t^+ - t$: since $|x-y| \le x+\beta$ on the support, $\int|\log|x-y||\,\mathrm{d}\omega_\beta = 2\int\log^+|x-y|\,\mathrm{d}\omega_\beta - U_\beta(x) \le 2\log(1+x+\beta) - \log(\beta/4)$. □

3.3. **Positivity of the energy of signed measures.**

Lemma 3.3. *Let ν be a finite signed Borel measure on $\mathbb{R}$ with $\nu(\mathbb{R}) = 0$ and $\iint |\log|x-y||\,\mathrm{d}|\nu|(x)\,\mathrm{d}|\nu|(y) < \infty$. Then*

$$-\iint \log|x-y|\,\mathrm{d}\nu(x)\,\mathrm{d}\nu(y) = \int_0^\infty \frac{|\hat\nu(\xi)|^2}{\xi}\,\mathrm{d}\xi \;\geq 0, \qquad \hat\nu(\xi) := \int e^{i\xi x}\,\mathrm{d}\nu(x),$$

with equality if and only if $\nu = 0$.

Proof. For $0 < \varepsilon < 1 < R$ and $a \in \mathbb{R}\setminus\{0\}$ put $K_{\varepsilon,R}(a) := \int_\varepsilon^R (\cos(a\xi) - e^{-\xi})\,\xi^{-1}\,\mathrm{d}\xi$.

Step 1: a uniform bound. We use three elementary estimates: (a) for $1 \leq A < B$, $|\int_A^B \cos t\,\frac{\mathrm{d}t}{t}| \leq 2/A \leq 2$ (integrate by parts); (b) for $0 < A < B \leq 1$, $\int_A^B \cos t\,\frac{\mathrm{d}t}{t} = \log\frac{B}{A} - \theta$ with $0 \leq \theta \leq \int_A^B \frac{t}{2}\,\mathrm{d}t \leq \frac{B^2}{4} \leq \frac14$, because $0 \leq 1-\cos t \leq t^2/2$; (c) $\int_\varepsilon^R e^{-\xi}\,\frac{\mathrm{d}\xi}{\xi} = \log\frac{1}{\varepsilon} + \theta'$ with $|\theta'| \leq 2$. Substitute $t = |a|\xi$, so that $\int_\varepsilon^R \cos(a\xi)\,\frac{\mathrm{d}\xi}{\xi} = \int_{\varepsilon|a|}^{R|a|} \cos t\,\frac{\mathrm{d}t}{t}$. If $\varepsilon|a| < 1 < R|a|$, then $K_{\varepsilon,R}(a) = -\log|a| + O(1)$. If $R|a| \leq 1$, then $K_{\varepsilon,R}(a) = \log R + O(1)$ and $0 < \log R \leq \log(1/|a|)$. If $\varepsilon|a| \geq 1$, then $K_{\varepsilon,R}(a) = -\log(1/\varepsilon) + O(1)$ and $0 < \log(1/\varepsilon) \leq \log|a|$. In all cases

$$|K_{\varepsilon,R}(a)| \leq \big|\log|a|\big| + 6 \qquad (0 < \varepsilon < 1 < R,\ a \neq 0). \tag{3.2}$$

Step 2: the pointwise limit. We have $K_{\varepsilon,R}(a) - K_{\varepsilon,R}(1) = \int_{\varepsilon|a|}^{\varepsilon} \cos t\,\frac{\mathrm{d}t}{t} + \int_R^{R|a|} \cos t\,\frac{\mathrm{d}t}{t}$. As $\varepsilon \to 0$ the first integral tends to $\log(1/|a|)$, by (b); as $R \to \infty$ the second tends to 0, by (a). Also $K_{\varepsilon,R}(1) \to k_0 := \int_0^\infty (\cos\xi - e^{-\xi})\xi^{-1}\,\mathrm{d}\xi$, which converges. Hence $K_{\varepsilon,R}(a) \to k_0 - \log|a|$. (In fact $k_0 = 0$, but we do not need this, because $\nu(\mathbb{R}) = 0$.) All limits as $\varepsilon \to 0$, $R \to \infty$ are taken along arbitrary sequences $\varepsilon_m \downarrow 0$, $R_m \uparrow \infty$.

Step 3. The integrability hypothesis forces $|\nu|\otimes|\nu|$ to give zero mass to the diagonal $\{x = y\}$. By (3.2) and dominated convergence along (ε_m, R_m) (with respect to $|\nu|\otimes|\nu|$, dominating function $|\log|x-y|| + 6$), and since $\nu(\mathbb{R}) = 0$,

$$\iint K_{\varepsilon,R}(x-y)\,\mathrm{d}\nu\,\mathrm{d}\nu \longrightarrow k_0\,\nu(\mathbb{R})^2 - \iint \log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu = -\iint \log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu.$$

On the other hand, the integrand $(\xi, x, y) \mapsto (\cos((x-y)\xi) - e^{-\xi})/\xi$ is bounded on $[\varepsilon, R]\times\mathbb{R}^2$, so by Fubini, $\iint \cos((x-y)\xi)\,\mathrm{d}\nu\,\mathrm{d}\nu = |\hat\nu(\xi)|^2$ and $\nu(\mathbb{R}) = 0$,

$$\iint K_{\varepsilon,R}(x-y)\,\mathrm{d}\nu\,\mathrm{d}\nu = \int_\varepsilon^R \frac{|\hat\nu(\xi)|^2}{\xi}\,\mathrm{d}\xi.$$

Letting $\varepsilon \to 0$, $R \to \infty$ (monotone convergence on the right) gives the identity.

Equality. If the integral vanishes, then $\hat\nu = 0$ on $(0,\infty)$ by continuity, hence on $\mathbb{R}$, because $\hat\nu(-\xi) = \overline{\hat\nu(\xi)}$ and $\hat\nu(0) = \nu(\mathbb{R}) = 0$. Then $\nu = 0$: for the Gaussian $G_t(x) = (2\pi t)^{-1/2}e^{-x^2/2t}$ we have $G_t(x) = \frac{1}{2\pi}\int e^{-t\xi^2/2}e^{-i\xi x}\,\mathrm{d}\xi$, so by Fubini $(G_t * \nu)(x) = \frac{1}{2\pi}\int e^{-t\xi^2/2}e^{-i\xi x}\hat\nu(\xi)\,\mathrm{d}\xi = 0$ for all x and $t > 0$; for bounded continuous ψ, $\int \psi\,\mathrm{d}\nu = \lim_{t\to0}\int (G_t * \psi)\,\mathrm{d}\nu = \lim_{t\to0}\int \psi\,(G_t * \nu)\,\mathrm{d}x = 0$ by dominated convergence, so $\nu = 0$. □

Remark 3.4. The positivity statement of Lemma 3.3 (but not the Fourier identity) is a special case of [2, Lemma 2.1]. Indeed, under the hypothesis of Lemma 3.3, the positive and negative parts $\nu^\pm$ of ν have equal mass and finite energy, and they satisfy $\int \log(1+|t|)\,\mathrm{d}\nu^\pm(t) < \infty$. For the last point, fix y outside a $|\nu|$-null set with $\int |\log|x-y||\,\mathrm{d}|\nu|(x) < \infty$, and use $\log(1+|x|) \leq \log(1+|y|) + \log 2 + \log^+|x-y|$. So $\nu^\pm \in M(\mathbb{C})$ in the notation of [2]. In [2] the energy is denoted $I(\mu) = \iint \log\frac{1}{|x-y|}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y)$ (not to be confused with our constant $I(p)$), $M(\Sigma)$ denotes the Borel measures of finite mass supported in Σ with finite energy and with $\int \log(1+|t|)\,\mathrm{d}\mu(t) < \infty$ [2, (1.12)], and the statement reads: "Lemma 2.1. Let $\mu, \nu \in M(\mathbb{C})$

with $\|\mu\| = \|\nu\|$. Then $I(\mu-\nu) \geq 0$, and $I(\mu-\nu) = 0$ if and only if $\mu = \nu$." The proof given there refers to the literature for both the compactly supported and the unbounded case. In the application below ($\mu = g_*\mu_0$ with $\mu_0 \in \mathcal{A}$ and $\nu = \mu^*$), both measures are positive, of mass 1, of finite energy, and satisfy $\int \log(1+x) < \infty$, so they lie in $M(\mathbb{C})$ and [2, Lemma 2.1] applies as well. Either reference suffices; we use Lemma 3.3.

3.4. **The explicit maximiser.** Throughout this subsection $c = c(p)$ and $\beta_t := 4ct^{-p}$ for $t \in (0,1]$. Thus μ^* is the mixture $\mu^*(B) = \int_0^1 \omega_{\beta_t}(B)\,\mathrm{d}t$ for Borel sets B. Put $U^*(x) := \int \log|x-y|\,\mathrm{d}\mu^*(y)$.

Proof of Theorem 1.3(i). Fix $x > 0$ and let $t_x := (4c/x)^{1/p}$, so that $x < \beta_t$ if and only if $t < t_x$. Then

$$\frac{\mathrm{d}\mu^*}{\mathrm{d}x} = \frac{1}{\pi}\int_0^{\min(1,t_x)} \frac{\mathrm{d}t}{\sqrt{x(\beta_t - x)}} = \frac{t_x}{\pi x}\int_0^{\min(1,1/t_x)} \frac{\mathrm{d}s}{\sqrt{s^{-p}-1}},$$

by the substitution $t = t_x s$, for which $\beta_t = xs^{-p}$. The substitution $v = s^p$ gives, for $0 \leq V \leq 1$,

$$\int_0^{V^{1/p}} \frac{\mathrm{d}s}{\sqrt{s^{-p}-1}} = \frac{1}{p}\int_0^V v^{\frac{1}{p}-\frac{1}{2}}(1-v)^{-\frac{1}{2}}\,\mathrm{d}v = \frac{1}{p}B\big(\tfrac{1}{p}+\tfrac{1}{2},\tfrac{1}{2}\big)\mathcal{B}_V\big(\tfrac{1}{p}+\tfrac{1}{2},\tfrac{1}{2}\big),$$

and $B(\frac{1}{p}+\frac{1}{2},\frac{1}{2}) = \sqrt{\pi}\,\Gamma(\frac{1}{p}+\frac{1}{2})/\Gamma(\frac{1}{p}+1) = 2I(p)$. With $V = \min(1, t_x^{-p}) = \min(1, x/4c)$ we obtain

$$\frac{\mathrm{d}\mu^*}{\mathrm{d}x} = \frac{(4c)^{1/p}x^{-1-1/p}}{\pi}\cdot\frac{2I(p)}{p}\cdot\mathcal{B}_{\min(1,x/4c)}\big(\tfrac{1}{p}+\tfrac{1}{2},\tfrac{1}{2}\big) = \mathcal{B}_{\min(1,x/4c)}\big(\tfrac{1}{p}+\tfrac{1}{2},\tfrac{1}{2}\big)\cdot\frac{\mathrm{d}\sigma_p}{\mathrm{d}x},$$

because $(4c)^{1/p} = \pi/(2I(p))$ by the definition of $c = c(p)$. Thus $\mu^* = \sigma_p$ on $[4c,\infty)$ and $\mu^* < \sigma_p$ (in density) on $(0,4c)$. Pulling back by $x = u^{-p}$, which maps σ_p to Lebesgue measure, gives the density ρ^* of μ_u^*; ρ^* is continuous and positive on $(0,\infty)$, equal to 1 exactly on $(0,u_s]$, and $u_s = (4c)^{-1/p} = 2I(p)/\pi$. The total mass is $\int_0^1 \mathrm{d}t = 1$. Hence $\mu_u^* \in \mathcal{A}$. □

Proof of Theorem 1.3(ii). By (3.1), for fixed $x > 0$, $\int_0^1 \int |\log|x-y||\,\mathrm{d}\omega_{\beta_t}(y)\,\mathrm{d}t \leq \int_0^1 \big(2\log(1+x+\beta_t) + |\log(\beta_t/4)|\big)\,\mathrm{d}t < \infty$, since $\log\beta_t = \log 4c - p\log t$. By Fubini, $U^*(x) = \int_0^1 U_{\beta_t}(x)\,\mathrm{d}t$. If $x \leq 4c = \beta_1$, then $x \leq \beta_t$ for all $t \in (0,1]$, and Lemma 3.2 gives $U^*(x) = \int_0^1 \log(ct^{-p})\,\mathrm{d}t = \log c + p$. For all $x > 0$, Lemma 3.2 gives $U_{\beta_t}(x) \geq \log(\beta_t/4)$, hence $U^*(x) \geq \log c + p$. □

Proof of Theorem 1.3(iii). The value. For $0 < s \leq t \leq 1$ we have $\operatorname{supp}\omega_{\beta_t} \subset [0,\beta_t] \subset [0,\beta_s]$, so by Lemma 3.2

$$\iint \log|x-y|\,\mathrm{d}\omega_{\beta_s}(x)\,\mathrm{d}\omega_{\beta_t}(y) = \int U_{\beta_s}\,\mathrm{d}\omega_{\beta_t} = \log\frac{\beta_s}{4} = \log c - p\log\min(s,t).$$

By (3.1), $\iint |\log|x-y||\,\mathrm{d}\omega_{\beta_s}\,\mathrm{d}\omega_{\beta_t} \leq 2\log(1+\beta_s+\beta_t) + |\log(\beta_t/4)|$, which is integrable over $(s,t) \in (0,1]^2$. Hence $\mathcal{L}(\mu^*)$ converges absolutely, and by Fubini

$$\begin{aligned}\mathcal{J}_p(\mu_u^*) = \mathcal{L}(\mu^*) &= \int_0^1\int_0^1 \big(\log c - p\log\min(s,t)\big)\,\mathrm{d}s\,\mathrm{d}t \\ &= \log c - p\int_0^1 2(1-m)\log m\,\mathrm{d}m = \log c + \frac{3p}{2} = \kappa_p.\end{aligned}$$

Optimality. Let $\mu \in \mathcal{A}$. If $\mathcal{J}_p(\mu) = -\infty$ there is nothing to prove, so assume $\mathcal{J}_p(\mu) > -\infty$ and let $\mu^x = g_*\mu \leq \sigma_p$. We first check that all energies below converge absolutely.

- $\mathcal{L}(\mu^x)$: the positive part of the integrand is integrable (Section 3.1), and the negative part is integrable because $\mathcal{L}(\mu^x) > -\infty$.

- $\mathcal{L}(\mu^*)$: shown above.
- $\mathcal{L}(\mu^x,\mu^*)$: using $|t|=2t^+-t$, $\log^+|x-y|\le\log(1+x)+\log(1+y)$ and part (ii),

$$\iint\big|\log|x-y|\big|\,\mathrm{d}\mu^*(y)\,\mathrm{d}\mu^x(x)\le\int\Big(2\log(1+x)+2\int\log(1+y)\,\mathrm{d}\mu^*(y)-(\log c+p)\Big)\,\mathrm{d}\mu^x(x)<\infty,$$

since $\mu^*\le\sigma_p$ and $\mu^x\le\sigma_p$ have finite logarithmic moments and mass 1.

Let $\nu:=\mu^x-\mu^*$, a finite signed measure on $(0,\infty)$ with $\nu(\mathbb{R})=0$. Since $|\nu|\le\mu^x+\mu^*$, the three bullets give $\iint|\log|x-y||\,\mathrm{d}|\nu|\,\mathrm{d}|\nu|<\infty$. Expanding bilinearly (all terms are finite),

$$\mathcal{J}_p(\mu)=\mathcal{L}(\mu^x)=\mathcal{L}(\mu^*)+2\int U^*\,\mathrm{d}\nu+\iint\log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu.$$

By Lemma 3.3 the last term is ≤ 0, with equality only if $\nu=0$. For the middle term, ν has mass 0, so $\int U^*\,\mathrm{d}\nu=\int(U^*-\log c-p)\,\mathrm{d}\nu$. By (ii) the integrand vanishes on $(0,4c]$ and is ≥ 0 on $(4c,\infty)$, where $\nu=\mu^x-\sigma_p\le 0$ by (i). Hence $\int U^*\,\mathrm{d}\nu\le 0$, and $\mathcal{J}_p(\mu)\le\mathcal{L}(\mu^*)=\kappa_p$, with equality only if $\nu=0$, that is, $\mu=\mu^*_u$. □

Remark 3.5. The only step of the paper that uses Lemma 3.3 (equivalently [2, Lemma 2.1]) is the optimality and uniqueness part of Theorem 1.3(iii). In the proof of Theorem 1.1 it is needed for the inequality $E_p\le\kappa_p$, which enters the upper bound. The lower bound uses only part (i) and the value $\mathcal{J}_p(\mu^*_u)=\kappa_p$.

Remark 3.6. For $p=1$ we have $c=\frac12$, $u_s=\frac12$ and $\rho^*(u)=\mathcal{B}_{\min(1,1/2u)}(\frac32,\frac12)$, which is Monien's density ρ quoted in Section 1.2. Indeed both equal 1 on $(0,\frac12]$. For $u>\frac12$ put $V=1/(2u)$; since $B(\frac32,\frac12)=\frac{\pi}{2}$, the substitution $s=\sin^2\phi$ gives

$$\mathcal{B}_V\big(\tfrac32,\tfrac12\big)=\frac{2}{\pi}\int_0^V\sqrt{\frac{s}{1-s}}\,\mathrm{d}s=\frac{2}{\pi}\big(\phi_V-\sin\phi_V\cos\phi_V\big),\qquad\sin\phi_V=\sqrt{V},$$

and $\phi_V=\arctan(1/\sqrt{2u-1})$, $\sin\phi_V\cos\phi_V=\sqrt{2u-1}/(2u)$. As a check, the two expressions agree at eleven points of $[0.1,10^4]$ in certified (Arb) arithmetic, to within $4.5\cdot10^{-74}$ (supplementary script `certified/c04_monien_density.py`).

4. Discretisation and the upper bound

In Sections 4 and 5 we write, for a finite $S=\{k_1<\cdots<k_N\}\subset\mathbb{N}$,

$$V(S):=\prod_{i<j}\big(k_i^{-p}-k_j^{-p}\big)^2,\qquad u_i:=k_i/n.$$

Since $k^{-p}=n^{-p}g(k/n)$,

$$\log V(S)=-pN(N-1)\log n+\sum_{i\ne j}\log|g(u_i)-g(u_j)|.\tag{4.1}$$

Lemma 4.1 (Discretisation)**.** *Let $M\ge1$. There is C_M such that for all $n\ge2$ and all $S=\{k_1<\cdots<k_N\}\subset\mathbb{N}\cap[1,Mn]$ with $N\le n$,*

$$\sum_{i\ne j}\log|g(u_i)-g(u_j)|\le n^2\mathcal{J}_p(\bar\mu_S)+C_Mn\log n,$$

where $\bar\mu_S\in\mathcal{A}_{N/n}$ has density 1 on $\bigcup_{k\in S}A_k$, $A_k:=[(k-1)/n,k/n]$.

Proof. For cells A, A' write $\langle F\rangle_{A\times A'} := n^2 \iint_{A\times A'} F(u,v)\,\mathrm{d}u\,\mathrm{d}v$ for the average of $F(u,v) := \log|g(u)-g(v)|$. Then $n^2\mathcal{J}_p(\bar\mu_S) = \sum_{i\neq j}\langle F\rangle_{A_{k_i}\times A_{k_j}} + \sum_i \langle F\rangle_{A_{k_i}^2}$. Recall that $|g'(u)| = pu^{-p-1}$ is decreasing.

Non-adjacent pairs. Let $k<l$ with $l\geq k+2$. For $u\in A_k$, $v\in A_l$ we have $g(u)-g(v)\geq g(k/n)-g((l-1)/n) =: D>0$. With $\Delta := g((l-1)/n)-g(l/n) \leq |g'((l-1)/n)|/n$ and $D\geq \frac{l-1-k}{n}|g'((l-1)/n)|$ (mean value theorem and monotonicity of $|g'|$),

$$\log\bigl(g(k/n)-g(l/n)\bigr) = \log(D+\Delta) \leq \log D + \frac{\Delta}{D} \leq \langle F\rangle_{A_k\times A_l} + \frac{1}{l-k-1}.$$

Adjacent pairs. Let $l=k+1$. The point value is $\log(g(k/n)-g((k+1)/n)) \leq \log(|g'(k/n)|/n)$. For $u\in A_k$, $v\in A_{k+1}$, $g(u)-g(v)\geq |g'((k+1)/n)|(v-u)$, so $\langle F\rangle_{A_k\times A_{k+1}} \geq \log|g'((k+1)/n)| - \log n + \vartheta_0$ with $\vartheta_0 := \int_0^1\int_1^2 \log(y-x)\,\mathrm{d}y\,\mathrm{d}x = 2\log 2 - \frac{3}{2}$. Since $|g'(k/n)|/|g'((k+1)/n)| = (1+1/k)^{p+1} \leq 2^{p+1}$, the point value exceeds the average by at most $(p+1)\log 2 - \vartheta_0$.

Diagonal cells. For $u,v\in A_k$, $|g(u)-g(v)|\geq |g'(k/n)||u-v|$, so $\langle F\rangle_{A_k^2} \geq \log|g'(k/n)| - \log n - \frac{3}{2}$, using $\iint_{[0,1]^2}\log|x-y| = -\frac{3}{2}$. For $k\leq Mn$, $|g'(k/n)|\geq pM^{-p-1}$.

Summation. For fixed i, the partners $j>i$ with $k_j\geq k_i+2$ have distinct values $k_j-k_i-1\geq 1$, so $\sum_j 1/(k_j-k_i-1)\leq 1+\log n$; there is at most one adjacent partner. Counting ordered pairs,

$$\sum_{i\neq j}\log|g(u_i)-g(u_j)| \leq n^2\mathcal{J}_p(\bar\mu_S) + 2N\bigl(1+\log n + (p+1)\log 2 + |\vartheta_0|\bigr) + N\bigl(\log n + \tfrac{3}{2} + (p+1)\log M + |\log p|\bigr),$$

which is of the stated form since $N\leq n$. □

The diagonal estimate is the only place where $S\subset[1,Mn]$ is used: an isolated point $k\gg n$ makes $\mathcal{J}_p(\bar\mu_S)$ very negative while contributing little to the point sum. Atoms beyond Mn are therefore treated separately in Theorem 4.3.

Corollary 4.2. *For $M\geq 1$ there is C'_M such that for $n\geq 2$ and $S\subset\mathbb{N}\cap[1,Mn]$ with $|S|=N\leq n$,*

$$\log V(S) \leq -pN^2\log N + E_pN^2 + C'_M n\log n.$$

Proof. For $N=0$ both sides are trivial. Otherwise, by (4.1), Lemma 4.1 and Lemma 3.1 with $\tau = N/n$,

$$\log V(S) \leq -pN(N-1)\log n + N^2\bigl(E_p - p\log(N/n)\bigr) + C_M n\log n,$$

and $-pN(N-1)\log n - pN^2\log(N/n) = -pN^2\log N + pN\log n$. □

Note that only the value of the supremum E_p enters here; no information about maximisers is used.

Theorem 4.3 (Upper bound)**.** *Let $w_k\geq 0$ with $W=\sum_k w_k<\infty$ and $\sigma_0 := \sum_k w_k\delta_{k^{-p}}$. Then*

$$\limsup_{n\to\infty}\frac{1}{n^2}\Bigl(\log D_n(\sigma_0) + pn^2\log n\Bigr) \leq E_p.$$

Proof. Fix $M\geq 2$. Each n-set $S\subset\mathbb{N}$ splits uniquely as $S = H\cup T$ with $H := S\cap[1,Mn]$ and $T := S\cap(Mn,\infty)$; let $m := |T|$ and $N := n-m = |H|$. Then $V(S) = V(H)V(T)\prod_{a\in H,b\in T}(a^{-p}-b^{-p})^2$.

Cross pairs. For $a \in H$ and $b \in T$, $0 < a^{-p} - b^{-p} < a^{-p}$. Since H consists of N distinct positive integers, $\prod_{a\in H} a \geq N!$, so the cross product is at most $(N!)^{-2pm}$.

Tail. All atoms b^{-p} with $b \in T$ lie in the interval $(0, (Mn)^{-p})$; hence

$$V(T) \leq (Mn)^{-pm(m-1)}, \qquad \sum_{|T|=m} w(T) \leq W^m.$$

Head. $\sum_{|H|=N} w(H)V(H) \leq W^N \sup_H V(H)$, and Corollary 4.2 bounds $\sup_H V(H)$.

By Lemma 2.1, $D_n(\sigma_0) \leq \sum_{m=0}^{n} T_m$, where

$$\log T_m \leq n \log^+ W - 2pm \log N! - pN^2 \log N + E_p N^2 - pm(m-1)\log(Mn) + C'_M n \log n.$$

Recall that $E_p = \kappa_p$ is finite (Theorem 1.3(iii)). Put $t := m/n$. If $N \geq 1$, use $\log N! \geq N \log N - N$, $\log N = \log n + \log(1-t)$ and $\log(Mn) = \log n + \log M$. The coefficient of $\log n$ is $-p(2mN + N^2 + m^2) = -pn^2$, and the remaining terms give

$$\log T_m + pn^2 \log n - E_p n^2 \leq n^2 \phi_M(t) + O_M(n \log n),$$

$$\phi_M(t) := -p(1-t)(1+t)\log(1-t) + 2pt(1-t) - E_p\, t(2-t) - pt^2 \log M.$$

Since $-(1-t)\log(1-t) \leq t$ on $[0,1)$, we get $\phi_M(t) \leq p(1+t)t + 2pt + 2|E_p|t - pt^2 \log M \leq At - pt^2 \log M$ with $A := 4p + 2|E_p|$. If $N = 0$ ($t = 1$), then directly $\log T_n \leq n \log^+ W - pn(n-1)\log(Mn) + C'_M n \log n$, and $-p \log M - E_p \leq A - p \log M$, so the same bound holds. The $O_M(n \log n)$ terms are uniform in m, and $\max_{t\in[0,1]}(At - pt^2 \log M) \leq A^2/(4p \log M)$. Summing the $n+1$ terms,

$$\limsup_{n\to\infty} \frac{1}{n^2}\Big(\log D_n(\sigma_0) + pn^2 \log n\Big) \leq E_p + \frac{A^2}{4p \log M}.$$

Let $M \to \infty$. □

5. The lower bound

Lemma 5.1. *Let φ be decreasing on $(0,\infty)$ and integrable near 0, and let $\lambda \leq$ Leb be a measure on $(0,\infty)$ of mass $\delta > 0$ with $\int |\varphi|\,\mathrm{d}\lambda < \infty$. Then $\int \varphi\,\mathrm{d}\lambda \leq \int_0^\delta \varphi(u)\,\mathrm{d}u$.*

Proof. $\int \varphi\,\mathrm{d}\lambda - \int_0^\delta \varphi = \int_{(0,\delta]} \varphi\,\mathrm{d}(\lambda - \mathrm{Leb}) + \int_{(\delta,\infty)} \varphi\,\mathrm{d}\lambda \leq \varphi(\delta)\big(\lambda((0,\delta]) - \delta\big) + \varphi(\delta)\lambda((\delta,\infty)) = 0$, because $\lambda - \mathrm{Leb} \leq 0$ and $\varphi \geq \varphi(\delta)$ on $(0,\delta]$, and $\varphi \leq \varphi(\delta)$ on (δ,∞). □

Theorem 5.2 (Lower bound)**.** *Let $w_k \geq c_1 k^{-s}$ for all k, with $c_1 > 0$, $s \in \mathbb{R}$, and let $\sigma_0 = \sum_k w_k \delta_{k^{-p}}$ (with $\sum w_k < \infty$, so that the moments are finite). Then*

$$\liminf_{n\to\infty} \frac{1}{n^2}\Big(\log D_n(\sigma_0) + pn^2 \log n\Big) \geq \kappa_p.$$

Proof. Fix $\eta > 0$.

Step 1: a compactly supported admissible approximant. By Theorem 1.3(i), μ_u^* has a continuous positive density $\rho^* \leq 1$ on $(0,\infty)$. For $U > 0$ let $\mu_U := \mu_u^*|_{(0,U]}$, of mass m_U. By the proof of Theorem 1.3(iii), $|\log|g(u) - g(v)||$ is $\mu_u^* \otimes \mu_u^*$-integrable, so by dominated convergence $\mathcal{J}_p(\mu_U) \to \mathcal{J}_p(\mu_u^*) = \kappa_p$ and $m_U \to 1$ as $U \to \infty$. Let $\theta := (\mu_U)_{1/m_U}$ (notation of Lemma 3.1). Then θ has mass 1, density $\rho^*(um_U) \in [d_U, 1]$ on its support $(0, U']$, where $U' := U/m_U$ and $d_U := \min_{(0,U]} \rho^* \in (0,1]$, and density 0 beyond. So $\theta \in \mathcal{A}$, and by Lemma 3.1

$$\mathcal{J}_p(\theta) = m_U^{-2}\mathcal{J}_p(\mu_U) + p \log m_U \geq \kappa_p - \eta$$

for U large enough, which we fix. Constants C_U and $O_U(\cdot)$ below depend on p and U only (in Step 4 also on c_1 and s).

Step 2: the quantile configuration. Let F be the distribution function of θ; it is continuous and strictly increasing on $[0,U']$. For $1 \le i \le n$ let $B_i = [l_i, r_i] := [F^{-1}(\frac{i-1}{n}), F^{-1}(\frac{i}{n})]$, $q_i := F^{-1}(\frac{i-1/2}{n})$, $k_i := \lceil nq_i \rceil$ and $u_i := k_i/n \in [q_i, q_i + \frac{1}{n})$.

- Since $\theta \le \mathrm{Leb}$ and $\theta(B_i) = \frac{1}{n}$: $|B_i| \ge \frac{1}{n}$, so $l_i \ge \frac{i-1}{n}$ and $r_i \ge \frac{i}{n}$; also $r_i - q_i \ge \frac{1}{2n}$ and $q_i - l_i \ge \frac{1}{2n}$. Hence $q_{i+1} - q_i \ge \frac{1}{n}$, so the k_i are distinct, $k_1 \ge 1$, and $u_j - u_i \ge \frac{1}{n}$ for $i < j$. Moreover, for $i \le n-1$, $u_i < q_i + \frac{1}{n} \le r_i + \frac{1}{2n} \le r_{i+1}$.
- Since the density is $\ge d_U$ on $(0,U')$: $|B_i| \le \frac{1}{d_U n}$, so $r_i \le \frac{i}{d_U n}$, and $u_j \le r_j + \frac{1}{2n} \le \frac{j+1}{d_U n}$.
- $S^* := \{k_1 < \cdots < k_n\} \subset [1, U'n + 1]$.

Step 3: energy comparison. Write $F(u,v) = \log|g(u) - g(v)|$ and, for blocks B, B', put $\langle F\rangle_{B\times B'} := n^2 \iint_{B\times B'} F \,\mathrm{d}\theta \,\mathrm{d}\theta$; this is an average, since $\theta(B_i) = \frac{1}{n}$. Then

$$n^2 \mathcal{J}_p(\theta) = \sum_{i\ne j} \langle F\rangle_{B_i\times B_j} + \sum_i \langle F\rangle_{B_i^2}.$$

For $i < j$, $u \in B_i$ and $v \in B_j$ we have $u \le v$ and

$$F(u,v) = -p\log u + h(u/v), \qquad h(r) := \log(1 - r^p),$$

with h decreasing on $(0,1)$. We show that $F(u_i,u_j) \ge \langle F\rangle_{B_i\times B_j} - \mathrm{err}_{ij}$ with $\sum_{i<j} \mathrm{err}_{ij} = O_U(n\log n)$. We use repeatedly that for $0 < r < 1$,

$$\min(1,p)(1-r) \le 1 - r^p \le \max(1,p)(1-r). \tag{5.1}$$

(a) *The term $-p\log u$.* The point value minus the average is $-p\langle \log(u_i/u)\rangle_{B_i}$, and $u_i \le r_{i+1}$, $u \ge l_i$ on B_i. For $i \ge 2$, $\log(r_{i+1}/l_i) \le (r_{i+1} - l_i)/l_i \le \frac{2/(d_U n)}{(i-1)/n}$, so the loss is at most $2p/(d_U(i-1))$. For $i = 1$, Lemma 5.1 (with $\varphi = \log(1/u)$ and $\lambda = \theta|_{B_1}$, $\delta = 1/n$) gives $\langle \log(1/u)\rangle_{B_1} \le n\int_0^{1/n} \log\frac{1}{u}\,\mathrm{d}u = \log n + 1$, and $\log u_1 \le \log r_2 \le \log\frac{2}{d_U n}$; so the loss is at most $p(\log\frac{2}{d_U} + 1)$. Summing over $j > i$ and then over i gives $O_U(n\log n)$.

(b) *The term h, for $j \ge i+3$.* On $B_i \times B_j$, $u/v \ge a := l_i/r_j$, so the average of $h(u/v)$ is at most $h(a)$. At the point, $u_i/u_j \le b := r_{i+1}/l_j$, and $b < 1$ because $l_j = r_{j-1} \ge r_{i+2} > r_{i+1}$; so $h(u_i/u_j) \ge h(b)$. The loss is at most $h(a) - h(b) = \log\frac{1-a^p}{1-b^p} \le \frac{b^p - a^p}{1-b^p}$. By (5.1), $b^p - a^p = b^p(1 - (a/b)^p) \le \max(1,p)(1 - a/b)$.

- If $b \le \frac{1}{2}$: $1 - b^p \ge 1 - 2^{-p}$, and $1 - \frac{a}{b} \le \big(1 - \frac{l_i}{r_{i+1}}\big) + \big(1 - \frac{l_j}{r_j}\big) \le \frac{2/(d_U n)}{(i+1)/n} + \frac{1/(d_U n)}{j/n} \le \frac{3}{d_U(i+1)}$. The loss is at most $C_U/(i+1)$.
- If $b > \frac{1}{2}$: by (5.1), $1 - b^p \ge \min(1,p)(1-b) = \min(1,p)\frac{l_j - r_{i+1}}{l_j} \ge \min(1,p)\frac{j-i-2}{n l_j}$, since the $j-i-2$ blocks $B_{i+2}, \dots, B_{j-1}$ each have length $\ge \frac{1}{n}$. Also $1 - \frac{a}{b} = \frac{b-a}{b} \le 2(b-a)$ and $b - a \le \frac{r_{i+1} - l_i}{l_j} + \frac{l_i}{r_j}\cdot\frac{r_j - l_j}{l_j} \le \frac{3}{d_U n l_j}$. The loss is at most $C_U/(j-i-2)$.

Summing, $\sum_{j\ge i+3}\big(C_U/(i+1) + C_U/(j-i-2)\big) = O_U(n\log n)$.

(c) *The term h, for $j \in \{i+1, i+2\}$.* The average of h is ≤ 0 since $h \le 0$. At the point, by (5.1), $1 - (u_i/u_j)^p \ge \min(1,p)\frac{u_j - u_i}{u_j} \ge \min(1,p)\frac{1/n}{(j+1)/(d_U n)} = \frac{\min(1,p) d_U}{j+1}$. So the loss is at most $\log(j+1) + C_U$, and there are at most $2n$ such pairs: $O_U(n\log n)$.

(d) *Diagonal blocks.* $|g(u) - g(v)| \le g(\min(u,v))$, so $\langle F\rangle_{B_i^2} \le p\langle \log(1/\min(u,v))\rangle_{B_i^2}$. For $i \ge 2$, $\min(u,v) \ge l_i \ge \frac{i-1}{n}$, giving $\le p\log\frac{n}{i-1}$. For $i = 1$ and $n \ge 1/d_U$ we have $B_1 \subset (0,1]$, so $\log\frac{1}{\min(u,v)} \le \log\frac{1}{u} + \log\frac{1}{v}$, and Lemma 5.1 gives $\le 2p(\log n + 1)$. In total $\sum_i \langle F\rangle_{B_i^2} \le p\big((n-1)\log n - \log(n-1)!\big) + 2p(\log n + 1) = O(n)$.

Combining (a)–(d),

$$\sum_{i\neq j}\log|g(u_i)-g(u_j)| = 2\sum_{i<j}F(u_i,u_j) \geq n^2\mathcal{J}_p(\theta) - \sum_i \langle F\rangle_{B_i^2} - O_U(n\log n)$$
$$\geq n^2\mathcal{J}_p(\theta) - O_U(n\log n).$$

Step 4: conclusion. By Lemma 2.1, $D_n(\sigma_0) \geq w(S^*)V(S^*)$. Here $\log w(S^*) \geq n\log c_1 - |s|\,n\log(U'n+1)$, and by (4.1) and Step 3,

$$\log D_n(\sigma_0) \geq -pn(n-1)\log n + n^2(\kappa_p - \eta) - O_U(n\log n).$$

Let $n\to\infty$ and then $\eta\to 0$. □

The lower bound uses Theorem 1.3(i) and the value $\mathcal{J}_p(\mu_u^*) = \kappa_p$, but neither Lemma 3.3 nor any optimality statement.

6. Proof of Theorem 1.1 and Corollary 1.2

Proof of Theorem 1.1. Pure powers. Let $\sigma_0 = \sum_k w_k\delta_{k^{-p}}$ with weights satisfying (1.4). By Theorems 4.3, 5.2 and 1.3(iii) ($E_p = \kappa_p$), $\log D_n(\sigma_0) = -pn^2\log n + \kappa_p n^2 + o(n^2)$. By Lemma 2.2(iii), the measure $\sum_k w_k\delta_{Ck^{-p}}$ has $\log D_n$ larger by $n(n-1)\log C = n^2\log C + O(n)$.

General atoms. Let f satisfy (1.3) and put $h(\xi) := C\xi^{-p}$. Since f' is continuous, $\xi^{p+1}|f'(\xi)|$ is bounded below by a positive constant, and $\xi^{p+1}f'(\xi)\to -pC<0$, we have $f'<0$ on $[1,\infty)$; f is bounded because $|f'(\xi)|\leq C'\xi^{-p-1}$ is integrable. The ratio $\rho = f'/h' = \xi^{p+1}f'(\xi)/(-pC)$ is bounded above and below by positive constants and tends to 1. All determinants involved are positive by Lemma 2.2(iv), since $w_k>0$ for all k and f, h are injective. Lemma 2.3 gives $\log D_n(\sigma) = \log D_n(\sum_k w_k\delta_{Ck^{-p}}) + o(n^2)$. □

Proof of Corollary 1.2. By Lemma 2.4, $\log D_n = n\log m_0 + 2\sum_{j<n}(n-j)\log b_j$, so

$$\frac{2}{n^2}\sum_{j=1}^{n-1}(n-j)\log(j^p b_j) = \frac{1}{n^2}\Big(\log D_n - n\log m_0 + 2p\sum_{j=1}^{n-1}(n-j)\log j\Big).$$

By Theorem 1.1 and Lemma 2.5, the bracket equals $-pn^2\log n + (\kappa_p + \log C)n^2 + pn^2\log n - \frac{3p}{2}n^2 + o(n^2) = \log\big(c(p)C\big)n^2 + o(n^2)$. □

Remark 6.1. Theorem 1.1 gives the error $o(n^2)$ only. For pure powers the proof gives no rate, because the rate of $\mathcal{J}_p(\theta)\to\kappa_p$ in Step 1 of Theorem 5.2 and the factor $A^2/(4p\log M)$ in Theorem 4.3 are not quantified. The numerics of Section 8 are consistent with an error $O(n\log n)$ for pure powers.

7. Hankel determinants of zeta values

Proof of Corollary 1.4. Since $2a+b>1$, every entry $\zeta(a(i+j)+b)$ with $i,j\geq 1$ is a convergent Dirichlet series, and

$$\zeta\big(a(i+j)+b\big) = \sum_{k\geq 1} k^{-(2a+b)}\big(k^{-a}\big)^{(i-1)+(j-1)}.$$

Re-indexing $i,j\mapsto i-1, j-1$, the determinant is $D_n(\sigma)$ for $\sigma = \sum_k k^{-(2a+b)}\delta_{k^{-a}}$. Apply Theorem 1.1 with $p=a$, $f(\xi)=\xi^{-a}$ (so $C=1$ and $\xi^{a+1}f'(\xi)\equiv -a$), and $w_k = k^{-(2a+b)}$, for which $\sum_k w_k<\infty$ and $w_k\geq k^{-s}$ with $s=2a+b$. The special cases follow from $c(1)=\frac12$ and $c(2)=\frac{\pi^2}{16}$. □

For example, $\frac{3a}{2} + \log c(a)$ equals 0.806853, 2.516871, 4.127795, 5.696954 (rounded to six decimals) for $a = 1, 2, 3, 4$ (Table 1).

Remark 7.1 ($a = 1$). Monien's $H_n^{(r)}[\zeta] = \det_{1\le i,j\le n} \zeta(i+j+r)$ is the case $a = 1$, $b = r$ of Corollary 1.4, so $\log H_n^{(r)}[\zeta] = -n^2(\log 2n - \frac{3}{2}) + o(n^2)$ for every real $r > -1$. For $r = 0, 1$ this is Monien's weak asymptotic form [12], read with an $o(n^2)$ error term, as the expansion from Zagier's numerics quoted there suggests; Monien's "$\sim$" by itself asserts only the leading term.

Remark 7.2 ($a = 2$: comparison with closed forms). Baricz and Štampach [1] evaluate Hankel determinants of the Rayleigh functions $\sigma_{2m}(\nu)$. (Statement numbers refer to the arXiv version of [1]; the closed forms we use are confirmed independently below.) Their Corollary 3 reads "For $n \in \mathbb{N}$, $\nu \notin -\mathbb{N}$, and $\ell \in \{0, 1\}$, one has $\det H_n^{(\ell)}(\nu) = 2^{-2n(n+\ell)} \prod_{k=1}^{2n+\ell-1} (\nu+k)^{k-2n-\ell}$", where $H_n^{(\ell)}(\nu)$ has (i, j) entry $\sigma_{2(i+j)+2\ell-2}(\nu)$; their Remark 5 gives $\det H_n^{(2)}(\nu) = 2^{-2n(n+2)}(n+1)(n+\nu+1)\prod_{k=1}^{2n+1}(\nu+k)^{k-2n-2}$; and their (22) reads $\sigma_{2n}(1/2) = \zeta(2n)/\pi^{2n}$. Taking $\nu = \frac{1}{2}$ and rescaling rows and columns by powers of π yields closed forms for $\det_{1\le i,j\le n} \zeta(2(i+j)+b)$ with $b = 2\ell - 2 \in \{-2, 0, 2\}$. For example, for $b = 0$,

$$\log \det_{1\le i,j\le n} \zeta\big(2(i+j)\big) = 2n(n+1)\log\frac{\pi}{2} + \sum_{k=1}^{2n}(k-2n-1)\log\big(k+\tfrac{1}{2}\big).$$

We verified these three closed forms against direct determinants for $1 \le n \le 30$ in certified (Arb) ball arithmetic, with relative difference at most $4 \cdot 10^{-152}$ (supplementary script `certified/c03_zeta_hankel.py`). The asymptotics of the closed form are elementary: with $m = 2n+1$, $\sum_{k<m}(m-k)\log(k+\frac{1}{2}) = \frac{m^2}{2}\log m - \frac{3}{4}m^2 + O(m\log m)$ (as in Lemma 2.5), which gives

$$\log \det_{1\le i,j\le n} \zeta\big(2(i+j)\big) = -2n^2\log n + \big(3 + \log\tfrac{\pi^2}{16}\big)n^2 + O(n\log n),$$

in agreement with Corollary 1.4 for $a = 2$, $b = 0$, and with a better error term. Numerically (floating-point; script `papers/paper1/checks/bs_check.py`), for $\ell = 0, 1, 2$ the quantity $(\log\det + 2n^2\log n)/n^2 - \kappa_2$ equals $-1.33\cdot10^{-4}$, $-3.48\cdot10^{-4}$ and $-5.63\cdot10^{-4}$ at $n = 10^5$, and $(n/\log n)$ times it lies between -4.90 and -1.15 for $n \in \{10^2, 10^3, 10^4, 10^5\}$. The closed forms for $\ell = 0$ and $\ell = 2$ differ from the case $\ell = 1$ only by the power of 2 ($2^{-2n(n+\ell)}$), the power of π (exponent $2\sum_{i\le n}(2i+\ell-1)$), the range $2n+\ell-1$ of the product (with $m = 2n+\ell$ in place of $2n+1$), and, for $\ell = 2$, the factor $(n+1)(n+\frac{3}{2})$. These changes alter the right-hand side only by $O(n\log n)$. Hence the same computation gives $-2n^2\log n + (3+\log\frac{\pi^2}{16})n^2 + O(n\log n)$ for $b = -2$ and $b = 2$ as well. Thus for $a = 2$, $b \in \{-2, 0, 2\}$, Corollary 1.4 (with error $O(n\log n)$) also follows from [1]; the asymptotic statement is not made there.

8. Numerical illustrations

The supplementary material is the repository

`https://github.com/speagleg/jacobi-asymptotics` (tag `v1.0-paper1` [placeholder: tag to be created at submission]).

It contains the scripts that produced the numbers below, together with their stored outputs, the Lean sources of the formalisation of Section 1.4 (directory `lean/`), and the certified (Arb ball arithmetic) checks cited in this paper (directory `certified/`). Script names below are paths in that repository.

Except where marked as certified, the computations below are floating-point and serve only as illustrations. The scripts were re-run for this version, and the outputs reproduce the stored ones exactly.

Constants and the maximiser. Table 1 lists $c(p)$, u_s and κ_p, together with a Monte Carlo estimate of $\mathcal{L}(\mu^*)$ obtained by sampling μ^* directly (independently of the closed-form value). The entries in the columns $c(p)$, u_s and κ_p, and the four values of $\frac{3a}{2} + \log c(a)$ quoted in Section 7, are certified to be correctly rounded (Arb, `certified/c01_constants.py`); the Monte Carlo column is floating-point. The density ratio $\mathrm{d}\mu^*/\,\mathrm{d}\sigma_p$ evaluated by quadrature equals 1 for $x \geq 4c$ to four digits (one exception at $p = 0.5$, $x = 4000c$, where the quadrature returns 0.9862; the closed form of Theorem 1.3(i) gives exactly 1 there) and is strictly below 1 for $x < 4c$.

p	$c(p)$	$u_s = 2I(p)/\pi$	κ_p	Monte Carlo $\mathcal{L}(\mu^*)$
0.5	0.408248	0.3750	−0.145880	-0.1442 ± 0.0021
1	0.500000	0.5000	0.806853	0.8054 ± 0.0027
1.5	0.566272	0.5798	1.681320	1.6818 ± 0.0034
2	0.616850	0.6366	2.516871	2.5208 ± 0.0042
3	0.689213	0.7132	4.127795	4.1281 ± 0.0058
4	0.738565	0.7628	5.696954	5.6884 ± 0.0075
6	0.801586	0.8235	8.778837	8.7781 ± 0.0109

TABLE 1. Constants of Theorem 1.3. Sources: `certified/c01_constants.py` (columns 2–4, certified) and `research/G2/checks/check_energy.py` (Monte Carlo column).

The averaged law. For $x_k = k^{-p}$ and $w_k = k^{-s}$ truncated to $k \leq K$, the recurrence coefficients b_j were computed in extended precision, and $S_n := \frac{2}{n^2}\sum_{j<n}(n-j)\log(j^p b_j) - \log c(p)$ is shown in Table 2; by Corollary 1.2, $S_n \to 0$. Doubling K changes the tabulated values by at most $5.4 \cdot 10^{-4}$ ($p = 1.5$; this is about 20% of the entry at $n = 1600$, which is therefore only indicative), $2.8 \cdot 10^{-5}$ ($p = 3$), $1.4 \cdot 10^{-6}$ ($p = 4$) and $7 \cdot 10^{-8}$ ($p = 6$). For $p = 0.5$ the atoms decay too slowly for the truncation used (K-doubling changes S_n by 10^{-2}), so that case is not illustrated. For the zeta weights of Corollary 1.4 with $(a, b) = (3, 1)$ and $(4, 1)$ the same statistic is $-0.5544, -0.2181, -0.0750$ and $-0.7452, -0.2922, -0.1004$ at $n = 25, 100, 400$ (script `research/G2/checks/check_hz.py`). The identification of $\det \zeta(a(i+j)+b)$ with D_n of $\sum_k k^{-(2a+b)}\delta_{k^{-a}}$ was checked to 12 digits for $(a, b) \in \{(3, 1), (4, 1), (3, 2)\}$ and $n \in \{4, 8, 12\}$.

p	s	K	$n = 25$	50	100	200	400	1600
1.5	1.5	50000	−0.03377	−0.02370	−0.01531	−0.00942	−0.00567	−0.00267
3	3	20000	−0.17266	−0.11244	−0.06962	−0.04161	−0.02423	−0.00782
4	4	6000	−0.27151	−0.17494	−0.10757	−0.06399	−0.03714	–
6	6	2000	−0.46811	−0.29944	−0.18324	−0.10863	–	–

TABLE 2. The averaged statistic S_n, floating-point.
Script: `research/G2/checks/check_averaged.py`.

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